

Deepak Mallampati

APPLIED AI / ML ENGINEER - LLM APPS, AGENTIC WORKFLOWS & RAG

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US-based · F-1 OPT (authorized now)

PROFESSIONAL SUMMARY

Applied AI / ML Engineer with 4+ years shipping production LLM applications, agentic workflows, and RAG systems end-to-end - prototype to production - across regulated verticals (healthcare, financial services, enterprise). Balances autonomous agentic architectures with deterministic pipelines, and owns AI quality through real evaluation frameworks (RAGAS, BERTScore, custom domain metrics): **40% hallucination reduction, 92% precision** over large document corpora, and **42% token reduction** via conductor-subagent orchestration. Deep Python + AWS (Lambda, SageMaker, Bedrock), vector search, and MLOps discipline. Translates ambiguous business requirements into shipped systems fast.

SKILLS

Applied AI / LLM Engineering • Agentic + Deterministic Pipelines • RAG & Retrieval Quality • AI Evaluation (RAGAS, BERTScore) • Hallucination Detection • Python • AWS (Lambda, SageMaker, Bedrock) • Vector Search • LangChain / LangGraph • MLOps & Monitoring

Generative AI & LLMs: Claude, GPT-4o/GPT-5, Gemini, LLaMA, AWS Bedrock | LangChain/LangGraph | RAG Architecture | Multi-Agent Systems | Prompt Engineering | Fine-tuning (LoRA, QLoRA, PEFT)

AI Evaluation & NLP: RAGAS, BERTScore, custom domain metrics | Hallucination detection | Retrieval-quality measurement | NER, Text Classification, Summarization | Drift Detection | A/B Testing

Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Document Chunking Strategies | Semantic Caching

ML & Data: PyTorch, TensorFlow, Keras | Classification, Regression, Anomaly Detection, Forecasting | Pandas, NumPy | Data engineering pipelines | Spark, Databricks, Kafka

Cloud & MLOps: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure, GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD (Jenkins, Azure DevOps) | MLflow, Weights & Biases

Languages: Python (primary), JavaScript/TypeScript, Java, SQL | FastAPI, React

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Architected production **agentic LLM systems** on a conductor-subagent orchestration framework - decomposing complex objectives into context-scoped subagents that execute autonomously, **reducing token consumption by 42%** while sustaining task accuracy across multi-step workflows.
- Built high-performance **RAG pipelines** over large-scale healthcare corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and **reduce hallucination by 40%** (92% precision) on domain-specific queries.
- Established a structured **LLM evaluation & monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to benchmark iterations pre-deployment and surface regressions before release.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, **reducing failed production runs by 60%**.
- Optimized inference cost and latency via prompt compression, semantic caching, and model routing, sustaining SLA targets while cutting per-query overhead.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed AI agents and backend services for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Evaluated GPT-4o, Claude Sonnet, and Gemini and optimized prompts, improving task success rates by **27%** on internal eval datasets and informing model-selection decisions.
- Delivered LLM applications supporting **100,000+ requests daily** across 20+ internal teams; engineered content safeguards reducing unsafe outputs by 42%.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a privacy-preserving ML pipeline on clinical data (k-anonymity, l-diversity, Laplace differential privacy), cutting re-identification risk from **3.38% to 0.32%** while retaining 99% of baseline predictive performance.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 with a curated instruction dataset for controllable long-form generation.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Built supervised **regression and classification** models for equipment-failure prediction, anomaly detection, and capacity forecasting, with end-to-end feature engineering on raw sensor/operational data.
- Fine-tuned BERT and RoBERTa for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

MangaLens Live on Chrome Web Store

A shipped Chrome extension delivering real-time AI translation of manga and webtoons across 29+ sites - requirement to shipped product, owned end-to-end (the forward-deployed pattern).

LLMs, translation pipeline, browser integration

career-ops Autonomous

An autonomous AI job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - agentic orchestration in production.

Agentic workflows, LLM, automation

Privacy-Preserving Clinical ML Research

A framework quantifying the privacy-utility tradeoff of k-anonymity, l-diversity, and differential privacy on sensitive patient data.

scikit-learn, differential privacy

EDUCATION

M.S. in Computer Science

2025

Kent State University, OH - GPA 3.70 / 4.0